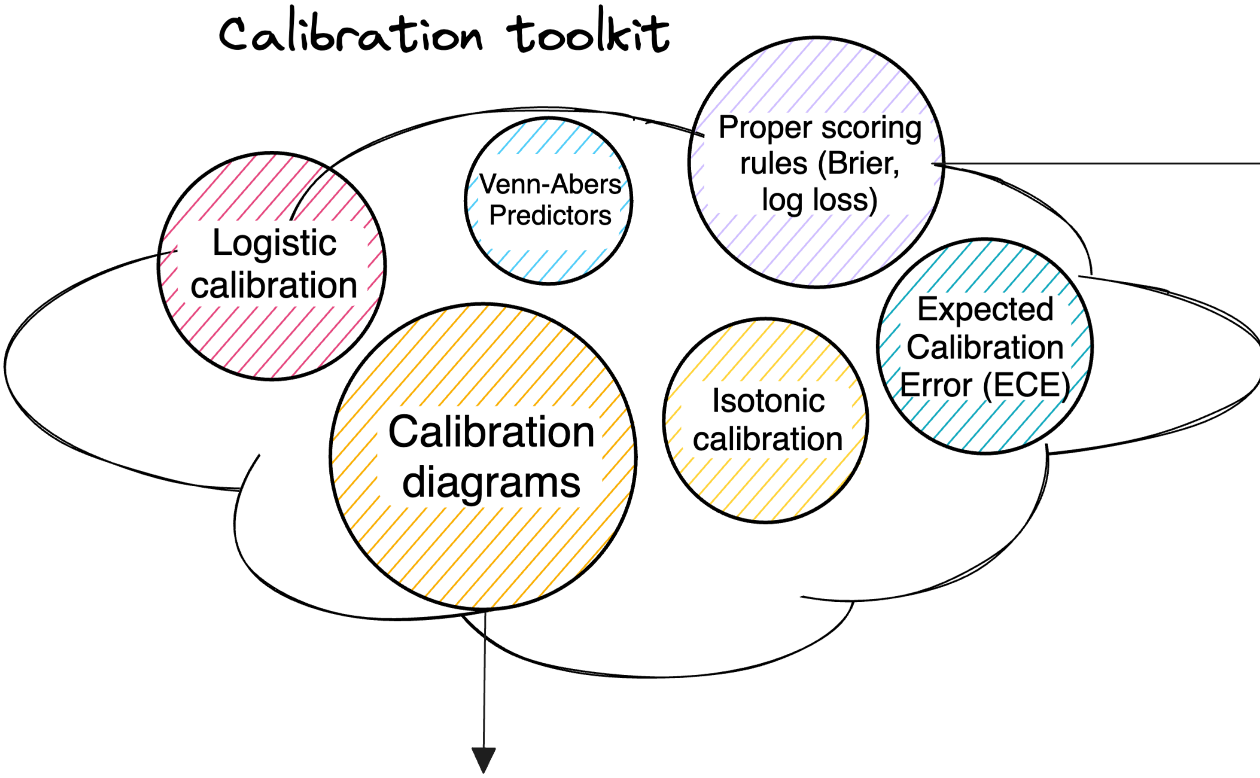
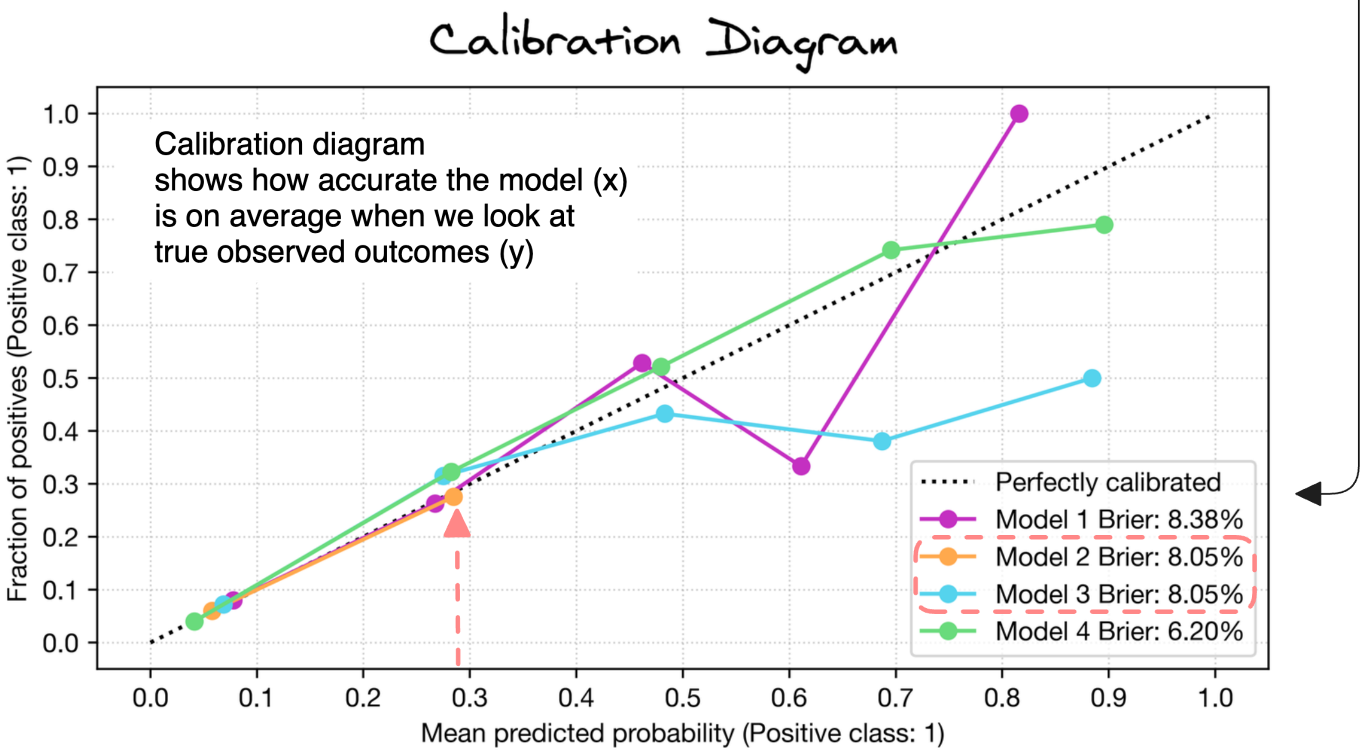
Calibration and model selection

Introduction

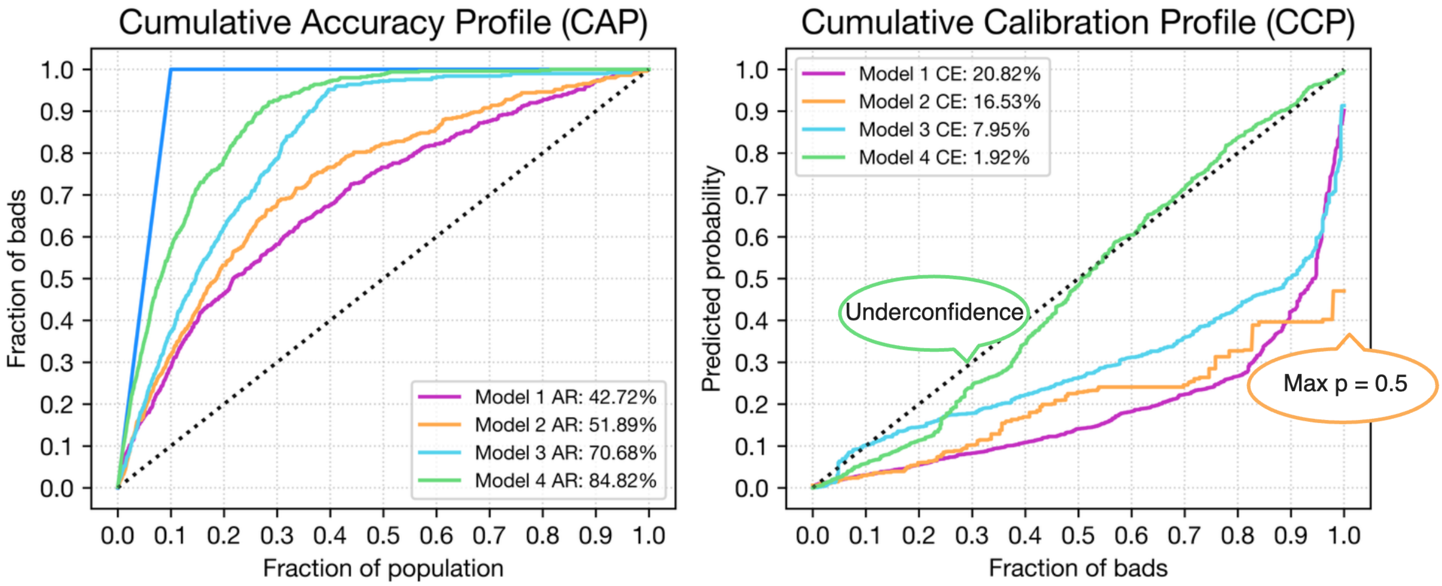




This model summary indicates that Models 2 and 3 have the same calibration accuracy, yet Model 2 has maximum prediction at 30% (under-confidence)

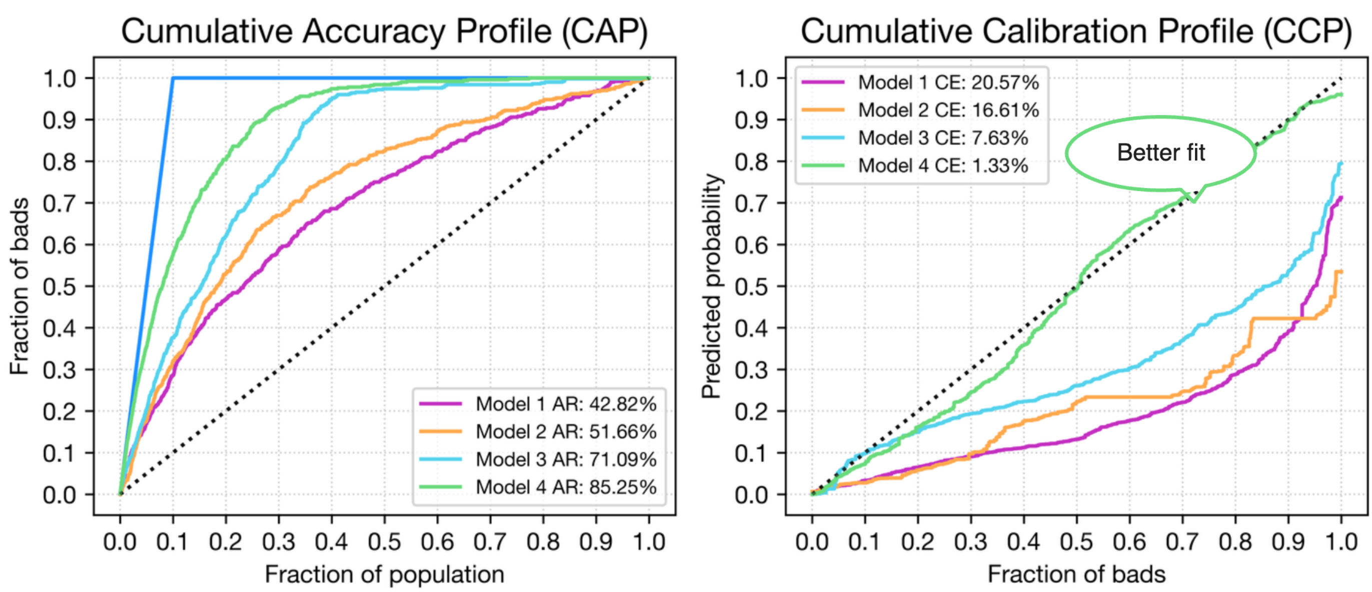
Calibration and model selection

XGBoost



Another way to look at calibration accuracy is to plot sorted predicted probabilities against the cumulative event rate where larger deviations are linked to both miscalibration and narrower range of predictions (around 0.5 for Model 2)

XGBoost (Calibrated)

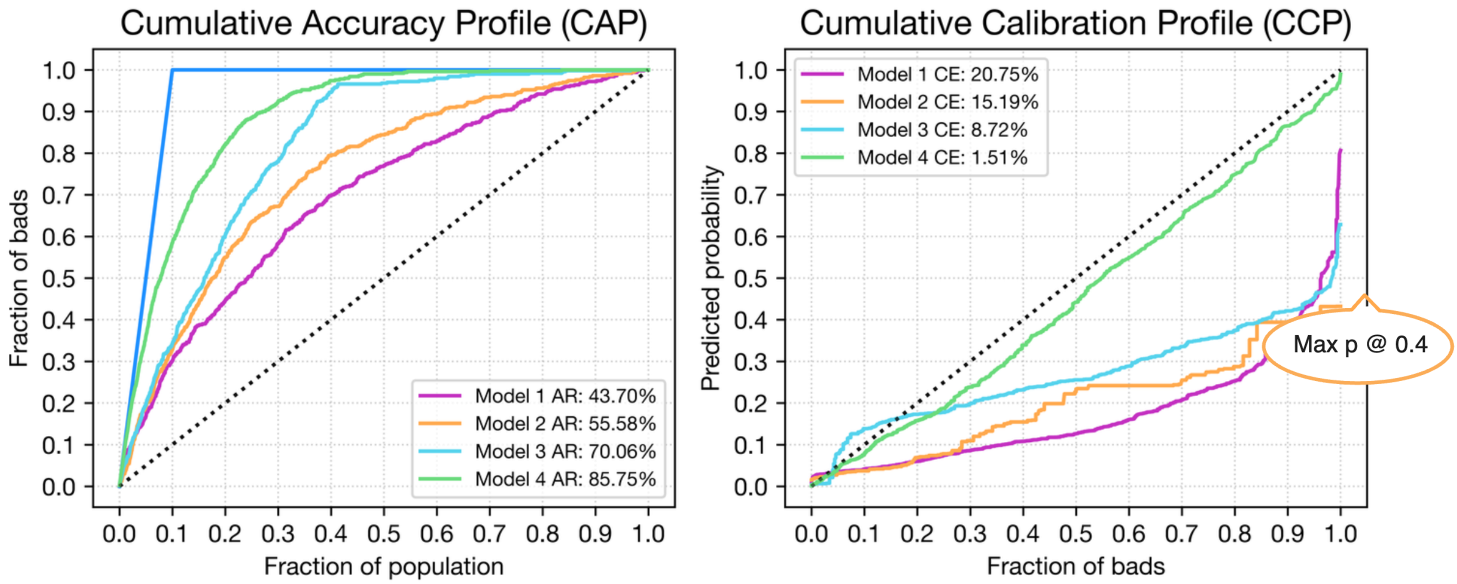


As we can see from the calibrated output (we isotonic calibration with cross-validation), no calibration technique can magically fix a bad model but it can improve a good one, at least which has a somewhat "well-calibrated" output

* CE = absolute error (like in ECE but without bins)

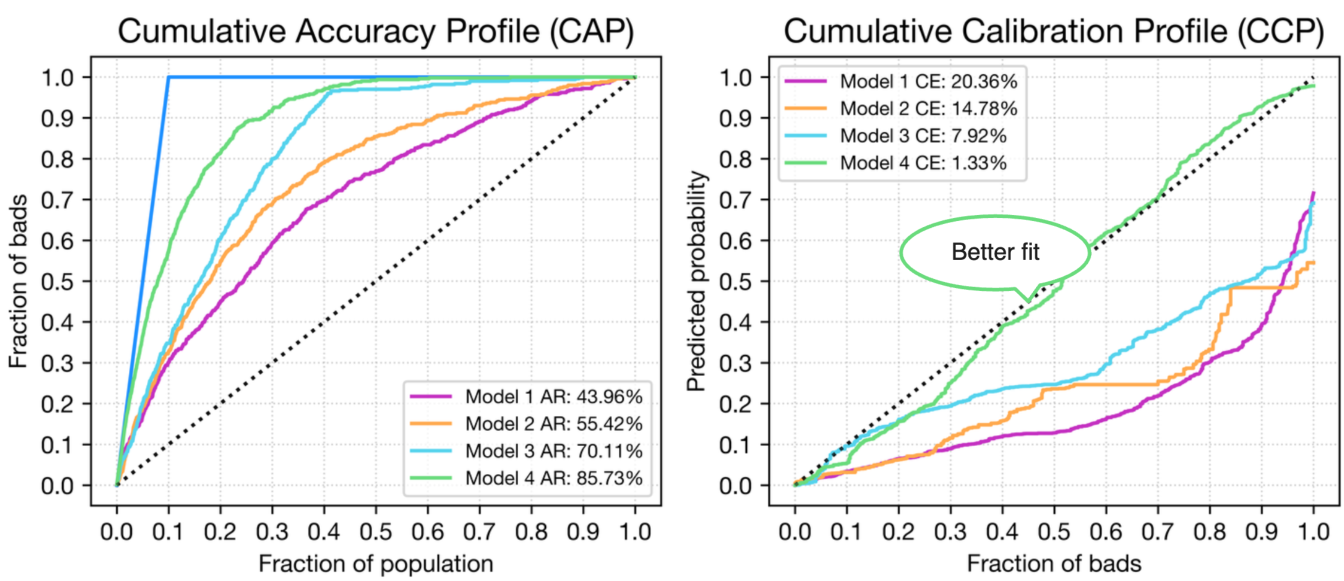
Calibration and model selection

CatBoost



We observe a quite similar picture for CatBoost, which has a slightly better raw calibration than XGBoost due to a different approach to constructing trees

CatBoost (Calibrated)

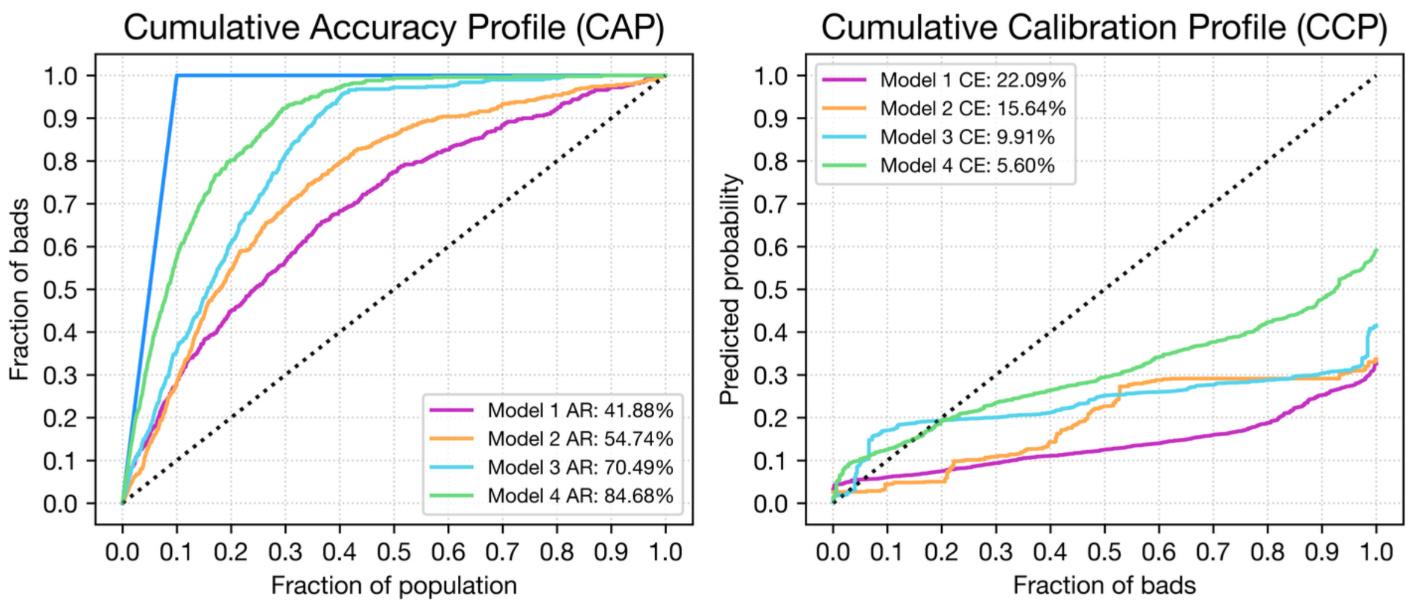


As with XGBoost that calibration can improve a good model's calibration a bit further, but it cannot remediate a poor raw output generated by weaker models as can be seen in the chart above

* CE = absolute error (like in ECE but without bins)

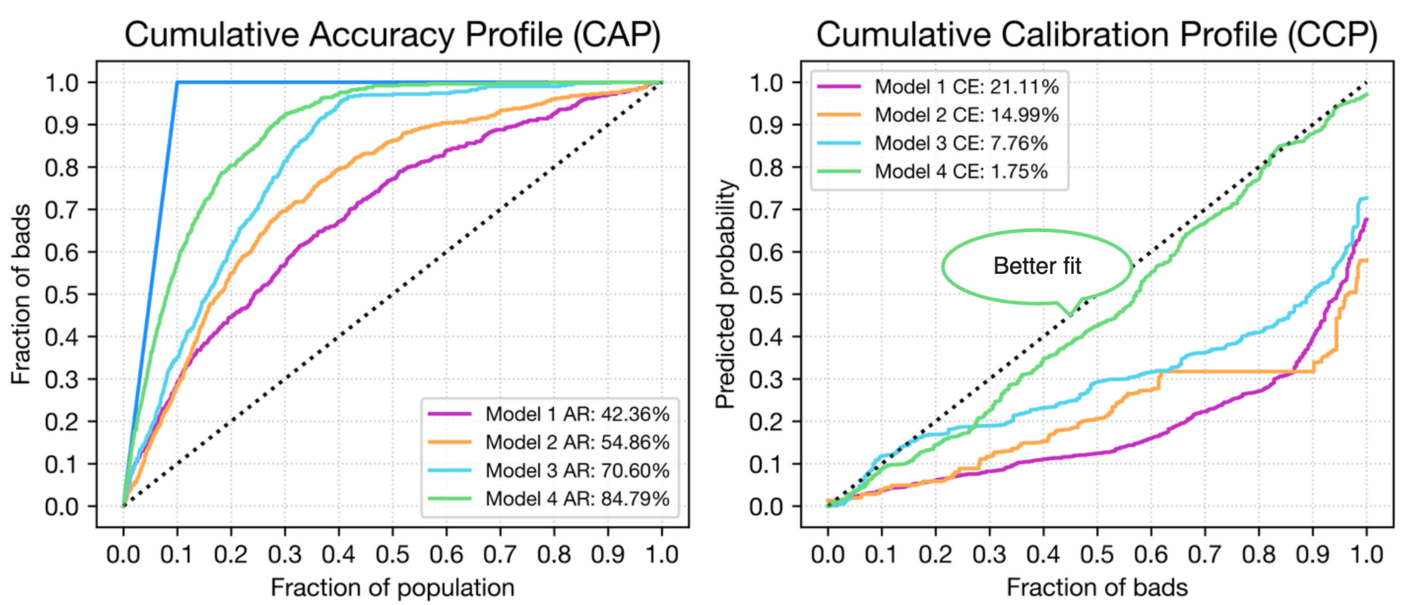
Calibration and model selection

Random Forest



It is quite well known that Random Forest (RF) produces ill-calibrated outputs. This could be driven by the fact that RF does not optimize any likelihood and is a data-mining approach

Random Forest (Calibrated)



Post-hoc calibration, as with other models, can slightly improve the accuracy of the model, but even with calibration RF models score worse in terms of calibration accuracy than other models (XGBoost, CatBoost)

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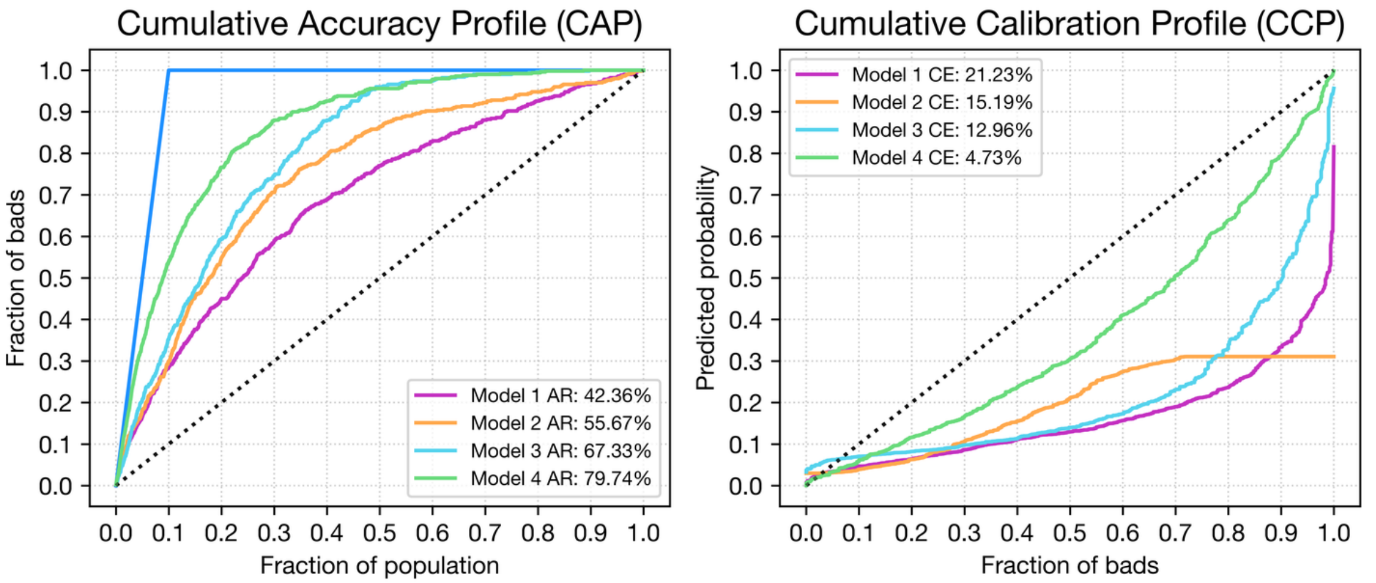
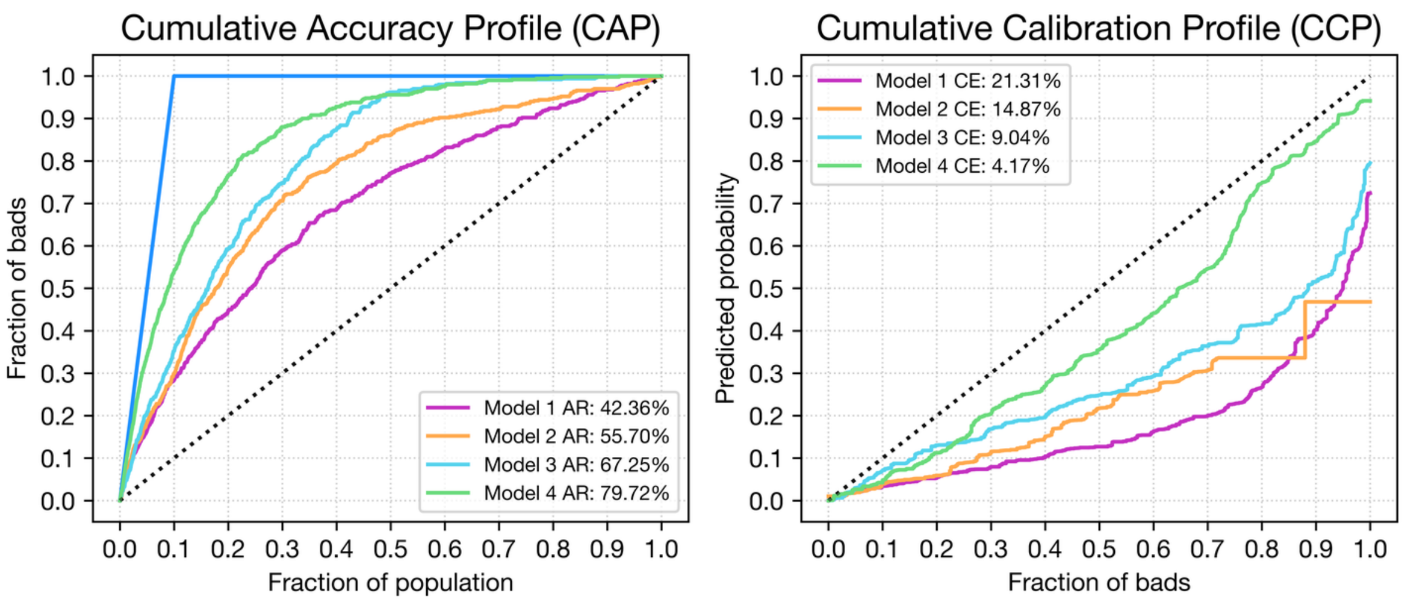
Notebook

Logistic Regression



A common misconception exists that logistic regression produces calibrated predictions. From the calibration curve you can see that it is not the case

Logistic Regression (Calibrated)



Even after calibrating, we still observe that the best model strongly deviates from the perfect line which is linked to the model's limited expressivity hence linear models may not be the most calibrated ones based on cumulative calibration analysis

* CE = absolute error (like in ECE but without bins)